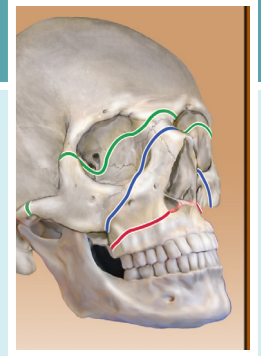


Deep Venous Thrombosis and Pulmonary Embolism

Jason Thompson, Marc M. DeHart



Bleeding in the athlete can result from injury or after orthopedic surgery. Hemostasis—the body’s process to stop bleeding—prevents what can be a life-threatening hemorrhage. Immobilization and hypercoagulable states can also induce clotting at improper sites (thrombosis). If the resultant clot dislodges and migrates (thromboembolism), it can lead to devastating tissue damage and organ failure. The most feared complication of that cascade is a fatal pulmonary embolism (PE). Although rare, it can occur after minor orthopedic surgeries, including arthroscopic procedures. The majority of clots cause few symptoms and are not recognized clinically. A reduction in peak aerobic performance may be the only hint of a clot in the elite athlete.¹ Chemical thromboembolic prophylaxis with anticoagulation therapy helps reduce the risk of thromboembolic events in the postoperative patient, but the use of these medications comes with significant risks of its own, as well as an added expense. Controversy surrounds the balance between the morbidity and mortality associated with thromboembolic events and the risks and costs of chemical prophylaxis.

NORMAL PHYSIOLOGY AND VIRCHOW TRIAD

Hemostasis and thrombosis are physiologic mechanisms of the coagulation system, platelets, endothelial cells, and the vascular wall. Following an injury, hemostasis is our body’s ability to form a blood clot and is usually followed by the dissolution of that clot as injured tissues repairs. Thrombosis can be considered “hemostasis in the wrong place and the wrong time.”² When a clot emerges in the arterial system, the subsequent loss of oxygenated blood can lead to stroke, myocardial infarction, and peripheral extremity necrosis. Clot formation within the venous system leads to local tissue congestion and decreased venous return, most often in the lower extremities. When the final destination of an embolus within the venous system is the lungs (PE), complications related to pulmonary infarction, abnormal gas exchange, and cardiovascular compromise may result.

In honor of Rudolph Virchow, who is responsible for coining the term *embolus*, “Virchow triad” describes the three primary influences on thrombus formation. *Endothelial damage* exposes collagen and triggers the extrinsic clotting cascade by activating platelets to perform their three primary functions—adhesion (sticking to damage endothelium), secretion (releasing thrombotic chemicals), and aggregation (combining platelets into a group). (Fig. 14.1). *Stasis* permits the bonds of protein clotting factors

and platelets to assemble and results from immobility (from postoperative/postinjury pain, cast, limb paralysis, stroke), increased blood viscosity (from cancer, estrogens, polycythemia), decreased inflow (from intraoperative tourniquets, vascular disease),³ and increased venous pressure (from venous scarring, varicose veins, heart failure).⁴ *Hypercoagulability* is the result of activation of the catalytic system of plasma proteins known as the coagulation cascade, whose main product—thrombin—converts soluble fibrinogen to insoluble fibrin. The biologic goal of this network of interdependent enzyme-mediated reactions is to limit hemorrhage at sites where damage occurs by rapidly stabilizing the initial platelet plug with insoluble fibrin.

Once formed, a thrombus has the ability to (a) undergo *dissolution* by the fibrinolytic system, (b) remain stationary with subsequent incorporation into the vein wall (*organization and recanalization*), (c) continue to grow (*propagation*), and/or completely or incompletely break free to travel downstream to imbed in the pulmonary vessels (*embolization*).⁴ Ninety percent of thrombi form in lower extremity veins, where those that form distal to the popliteal space occur in smaller veins of the calf and pose essentially no clinical threat because these typically dissolve spontaneously. However, a thrombus formed in the larger-diameter veins within the pelvis and proximal thigh are associated with increased risk of embolism.

PREOPERATIVE THROMBOEMBOLIC RISK FACTORS

Thrombophilia is the predisposition to venous thromboembolism disease (VTE) and is caused by inherited (primary) and acquired (secondary) factors. Primary hypercoagulability is often a result of genetic mutations, which may lead to an abnormal quantity or quality of protein clotting factors. Screening for prothrombotic defects has not been shown to be effective in selecting a strategy for thromboembolic prophylaxis.⁵ The quality and quantity of the protein clotting factors can be a consequence of either malproduction or autoimmune alteration or destruction of these important factors. Clinical factors of secondary hypercoagulability are commonly found in orthopedic patients and play a significant role in the perioperative management of thrombosis. These classic, sometimes modifiable, preoperative risk factors for VTE include, but are not limited to, previous VTE, malignancy, pregnancy, age older than 40 years, obesity, smoking, peripheral vascular disease, and oral contraceptive (and/or estrogen) use.

Abstract

Thromboembolic disease is a common complication of orthopedic surgery patients. This chapter includes a review of the basic science of hemostasis and discusses some of the sports-specific areas of interest for clinical practice.

Keywords

thromboembolism
blood clot
pulmonary embolism
venous thrombosis
hemostasis
effort thrombosis

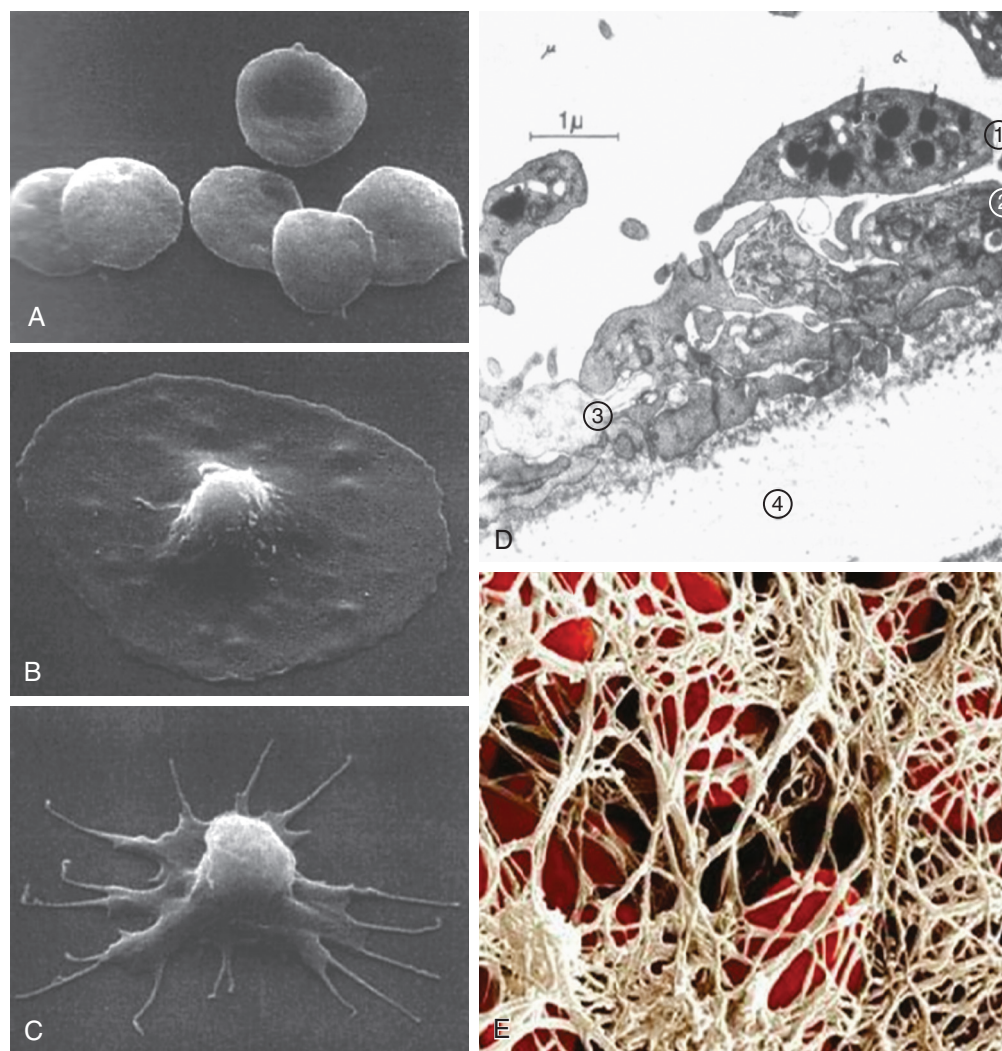


Fig. 14.1 (A) Scanning electron micrograph (SEM) of free platelets. (B) SEM of platelet adhesion. (C) SEM of platelet activation. (D) Transmission electron micrograph (TEM) of aggregating platelets. 1, Platelet before secretion. 2 and 3, Platelets secreting contents of granules. 4, Collagen of endothelium. (E) SEM of fibrin mesh encasing colored red blood cells. (Platelet EM courtesy James G. White, MD, Regents' Professor, Department of Laboratory Medicine and Pathology, University of Minnesota School of Medicine.)

Studies have found an association of high-altitude locations with a two to nine times greater rate of VTE in both young health populations and in autopsy records.^{6–8} At high altitude, hypoxia, dehydration, hemoconcentration, low temperature, and use of constrictive clothing, as well as enforced stasis due to severe weather, would support the occurrence of thrombotic disorders.⁹ Ascending from a low elevation to an elevated altitude (4000 feet above sea level) initially causes an acute hypoxic ventilation period, followed by a respiratory system response where the lungs increase ventilation (minutes to hours). Higher altitudes stimulate renal and hepatic erythropoietin production that then stimulates erythropoiesis with an increase in size and concentration of red cells. This erythropoietin process is most active 2 days after introduction to the elevation but usually returns to sea level measurements after approximately 3 weeks at higher elevation. Physiologic changes occur to acclimatize over the course

of several years with sustained hyperventilation while at higher elevation. Changes occur to the oxygen-hemoglobin dissociation curve, pulmonary circulation, cardiac function, fluid hemostasis, and hematologic components.¹⁰ The relative risk of VTE after knee arthroscopy in residents of areas 4000 feet above sea level is as high as 3.8 times that of similar patients residing in and having their operations in areas at sea level (but the risk is still very low, <.5%).¹⁰

Larger and longer operations on more proximal joints of the lower extremity have higher risks of clinically significant VTE. Careful individualized risk stratification may help identify patients at highest risk for VTE. Patients who have a history of VTEs and/or malignancy, or two or more of the classic risk factors, are at an increased risk of postoperative thromboembolic disease and may be considered for chemoprophylaxis for arthroscopy and distal procedures.¹¹

PERIOPERATIVE DECISION OF THROMBOEMBOLIC PROPHYLAXIS

The necessity of prophylaxis is based on balancing the risks of developing venous thrombosis and PE against the cost and dangers of prophylaxis. Operative factors under control of the surgeon can alter the risk of developing a VTE by influencing the size, location, and frequency of thrombosis. These perioperative factors include duration of operative and tourniquet time, complexity of the surgery, extremity paralysis, and length of postoperative extremity immobilization. Sports medicine generally involves less invasive procedures on younger and healthier patients who are often motivated to return to function at an accelerated pace. Despite a population with low thromboembolic risk, fatal PEs can occur and command considerable attention.¹²

The large volume of hip and knee arthroplasty that share elevated risks of VTE have provided a rich opportunity to study thromboembolic disease. Despite averaging a risk of deep venous thrombosis (DVT) of approximately 40% to 50%, the incidence of symptomatic/fatal PE is so low it is impractical to measure, and no study has had sufficient power to demonstrate a significant decrease. To provide statistically significant results, studies measure distal and proximal DVTs ("total DVT") as efficacy end points for prophylaxis. Total DVT is a high-frequency surrogate outcome that is practical to objectively measure but may not be clinically relevant. Experts question whether adequate data are available to say whether DVTs cause all PEs.^{13,14} The majority (79%) of studies on thromboprophylaxis after total joint arthroplasty were sponsored by industry, and evidence of potential bias exists.¹⁵ However, these studies include some of the highest level of evidence available in orthopedic surgery and are used by panels that create clinical practice guidelines and review comparative effectiveness. Most studies are powered to show efficacy in reducing total DVT. It takes a much smaller study group to statistically demonstrate a decrease in a high-frequency *but less clinically relevant* event such as total DVT (risks ~40%) than to detect an increase of a clinically relevant, low-frequency event such as bleeding (risks ~1% to 5%).^{13,16} Large prospective randomized trials frequently exclude patients who have increased bleeding risks, yet these patients suffer equally from arthritis and are often candidates for arthroplasty. Bleeding is a predictable complication of surgery and the most common complication of the use of anticoagulants. Higher rates of bleeding are seen when more effective prophylactic drugs are used and when drugs with more rapid onset of action are used closer to the time of surgery.¹⁷ Potent anticoagulation is not associated with a reduced rate of PE¹⁸ or mortality rate, nor is it associated with a reduction in the proportion of deaths related to PE.¹⁹ Observational studies show higher infection rates (odds ratio [OR] 1.5) with strict compliance with the use of strong chemoprophylaxis.²⁰ Modern arthroplasty literature finds wound-related complications, bleeding, hematoma formation, persistent wound drainage, and periprosthetic joint infection are reduced with the use of aspirin when compared with more powerful anticoagulants.¹⁴

Clinical practice guidelines, comparative effectiveness research, and consensus opinions can serve as a convenient starting point for surgeons and hospitals to make daily care decisions based

on current evidence. The orthopedic community has resisted a single standard regimen in favor of individualized judgments based on patient's clinical information. Understanding the recommendation guidelines creates opportunities for improved patient care (Table 14.1 Guidelines). Mechanical methods of prophylaxis have low bleeding risks and include aggressive range of motion, early weight-bearing activity, and the use of venous foot pumps and intermittent pneumatic compression hose.²¹ Use of mechanical devices is encouraged in patients at high risk for bleeding and as adjuncts with other prophylactic techniques. When VTE risks are substantial and bleeding risks allow, chemoprophylaxis is supported for patients undergoing major orthopedic surgeries, including operative fixation of pelvic fractures, hip fractures, polytrauma, and joint replacement of the hip and knee. Common VTE prophylactic and treatment medications are listed in Table 14.2 with their mechanisms of action (see Table 14.2 Prophylaxis Drugs).

The vast majority of sports medicine procedures have minimal thromboembolic risk (Table 14.3 DVT Incidence). Literature reviews on routine VTE prophylaxis with stronger anticoagulants suggest adverse events are more common in the intervention group and find no strong evidence for DVT prophylaxis for routine patients undergoing arthroscopy.²² Because the rates of symptomatic DVT and PE are so low following arthroscopic cases,^{17,23} guidelines recommend against routine use of chemoprophylaxis with the exception of using mechanical prophylaxis in the form of early mobilization. Despite the frequent need for immobilization and/or casting, smaller procedures such as distal fractures, tendon ruptures, and foot and ankle surgeries do not raise concern for an increased risk of VTE in the average patient. A large study that examined clinically symptomatic VTE in more than 45,000 patients undergoing foot and ankle surgery found the rate of VTE was 1%.²⁴ In one of the rare adequately powered multicentered randomized controlled studies of symptomatic thromboembolic disease in patients undergoing knee arthroscopy and a similar study in patients using casts concluded prophylaxis for these patients was not effective.²⁵ For sports patients with multiple risk factors undergoing long or complicated procedures, chemoprophylaxis may be effective and can be used when bleeding risks allow.

DIAGNOSIS AND CLINICAL MANIFESTATIONS: IMAGING AND LABORATORY FINDINGS

Diagnosis of venous thrombosis relies on a patient's history because there is no single physical exam finding that is diagnostic. Although VTE can cause leg pain, tenderness, swelling, or a palpable cord, most are asymptomatic. Homans sign (calf pain with foot dorsiflexion when the knee is flexed) and Moses' sign (pain with calf compression against the tibia) are known as the "classic" physical exam tests for a DVT. However, venographic evidence of concurrent DVT is found in fewer than 50% of patients who are positive for these classic signs of DVT (Fig. 14.2, upper panel).²⁶ Even with venographic evidence of VTE, only 50% of those patients will display the classical clinic signs.²⁷ Most orthopedic patients with a PE are asymptomatic. When emboli create large occlusions of the pulmonary vasculature, it

TABLE 14.1 Guidelines on Prevention of Venous Thromboembolism Disease in Hip and Knee Arthroplasty

AAOS Guideline and Grade of Recommendation: !Strong, +Moderate, ~Weak, *Consensus, ?Inconclusive		Notes From Other Guidelines: ¹ACCP, ²NICE, ³AHRQ
!	No "screening" duplex US	
+	Hx normal risks of VTE and bleeding use drugs <i>and/or</i> IPC	
+	D/C platelet inhibitors preop (aspirin, clopidogrel, prasugrel)	Discuss with medical team, stop 1 week prior ²
+	Neuraxial anesthesia to decrease bleeding (No effect on VTE)	Caution with drugs and neuraxial ¹ Wait 12 h after drugs ²
~	Ask history of previous VTE	
*	Hx of VTE get IPC <i>and</i> Drugs	DRUGS: LMWH, fondaparinux, dabigatran, rivaroxaban, VKA, aspirin ¹
*	Ask history of bleeding disorder (hemophilia) and active liver disease	
*	Hx of bleeding disorder (hemophilia or active liver disease) <i>only</i> IPC	If bleeding risk: IPC or nothing ¹ If bleeding risk > clotting risk: IPC ²
*	Discuss duration with patient	≥10 days, consider 35 days ¹
*	Early mobilization	
?	Assess other clotting risk factors	
?	Assess other bleeding risk factors	
?	No one technique optimal	Drugs and IPC ^{1,2} D/C drugs when TKA mobile ²
?	IVC filter	If VTE risks high and contraindication to prophylaxis ²
Guideline Title		Source
2011 AAOS Preventing Venous Thromboembolic Disease		http://www.aaos.org/research/guidelines
2012 ACCP Prevention of VTE in Orthopedic Surgery Patient		http://journal.publications.chestnet.org/
2012 AHRQ VTE Prophylaxis in Orthopedic Surgery, CER 49		http://effectivehealthcare.ahrq.gov/
2010 NICE Reducing the Risk of VTE in patients admitted to hospital		http://guidance.nice.org.uk/CG92/

D/C, Discontinue; IPC intermittent pneumatic compression; IVC, intra-vena caval; US, ultrasound; VKA, vitamin K antagonist; VTE, venous thromboembolism.

AAOS. Guideline on Preventing Venous Thromboembolic Disease in Patients Undergoing Elective Hip and Knee Arthroplasty. *AAOS Clinical Practice Guidelines (CPG)*; 2011. http://www.aaos.org/research/guidelines/VTE/VTE_guideline.asp/. Accessed June, 20, 2012.

AHRQ. Comparative Effectiveness Review, Number 49: Venous Thromboembolism Prophylaxis in Orthopedic Surgery; 2012. http://effectivehealthcare.ahrq.gov/ehc/products/186/992/CER-49_VTE_20120313.pdf/. Accessed July 20, 2012.

NICE. CG92 Venous thromboembolism—reducing the risk: NICE guideline; 2010. <http://guidance.nice.org.uk/CG92/NICEGuidance/pdf/English/>.

can lead to right heart failure and hypoxemia. Healthy individuals may remain asymptomatic if an occlusion obstructs less than 60% of the pulmonary circulation.²⁸ The most common presenting symptoms in patients diagnosed with PE on pulmonary angiogram are chest pain (typically pleuritic) and sudden onset of shortness of breath (dyspnea). Objective physical exam findings seen in greater than 50% of patients are tachypnea (>20 breaths/min) and crackles with lung auscultation.²⁸ Massive *saddle emboli* block all cardiopulmonary circulation and can cause immediate death.

The nonspecific nature of the signs and symptoms of VTE can be confused for other diagnoses or even easily dismissed, and therefore these complaints demand a high clinical suspicion in patients particularly at high risk. There is no ideal objective test for VTE, but contrast venography, duplex compression ultrasound, spiral computed tomography (CT) venography, CT pulmonary angiography, ventilation-perfusion (V/Q) scintigraphy, and D-dimer levels, usually used in combination, can be useful. Duplex ultrasound is the most practical diagnostic tool for lower extremity DVTs because it is inexpensive, noninvasive, and easily administered at the patient's bedside. Assessing blood flow and compressibility of the vein with real-time B-mode Doppler

ultrasound is more than 95% sensitive and specific for detecting proximal DVTs.²⁶ When a DVT is suspected, an urgent outpatient screening is indicated where duplex ultrasound serves as the best first line test in a stable patient (see [Fig. 14.2, lower panel](#)).²⁹ All guidelines agree that routine screening with duplex ultrasound at discharge from hospital is not cost effective and is not recommended.^{16,30} If nearby wounds, burns, or the presence of a cast or when DVT of the pelvis or inferior vena cava vessels are suspected, CT and magnetic resonance venography have shown good sensitivity and specificity.³¹

A postoperative patient who presents with chest pain, dyspnea, or cardiovascular collapse raises the clinical suspicion of PE; the diagnostic workup, provided that the patient is stable, should include a chest radiograph, electrocardiogram (ECG), and determination of an arterial blood gas (ABG) value. The plain film of the chest will often have subtle and nonspecific findings, whereas the ECG findings in patients with a PE potentially show sinus tachycardia, T-wave inversion, and ST abnormalities; however, these are not diagnostic alone. The tachypnea of a PE causes hyperventilation, which is represented by hypocapnia (low serum carbon dioxide level) in the ABG.³² D-dimer levels are often not helpful in the orthopedic setting because the trauma from

TABLE 14.2 Common Drugs for Venous Thromboembolism Disease Prophylaxis/Treatment

COMMON DRUGS FOR THROMBOEMBOLIC PROPHYLAXIS/TREATMENT				
Drug	Mechanism of Action	Complications	Notes	Reversal
Aspirin ^a	Irreversibly blocks platelet COX-1 production of TXA ₂	Bleeding & GI issues Rare allergies	Least effective chemical agent	None
Heparin ^b	Binds AT to inactivate Xa & II (thrombin) and IXa, Xla, Xlla	Bleeding HIT and osteoporosis	Must monitor: aPTT or anti-Xa activity	Protamine: 1 mg IV per 100 U UFH
LMWH ^b	Bind AT to inactivate Xa, & IIa (thrombin)	Bleeding Rare HIT	Monitor anti-Xa activity if: BMI >50 or CrCl <30	Protamine: 1 mg IV per 1 mg LMWH
Fondaparinux ^c	Binds AT to selectively inactivate only Xa	Bleeding	Avoid in: thin <50 kg, elderly >75, CrI: CrCl <30	No antidote
VKA ^d Coumadin	Blocks γ -carboxylation of: factors II, VII, IX, X	Bleeding Skin necrosis Fetal Warfarin syndrome	Oral good for long term therapy Multiple reactions: diet, drugs, disease	Vitamin K1 (phytonadione): FFP 8–10 mL/kg
Dabagatran ^d	Direct thrombin inhibitor	Bleeding	Oral	Idarucizumab (monoclonal Ab) ^e : 5 g IV
Rivaroxaban ^d	Direct factor Xa inhibitor	Bleeding	Oral Avoid if on antifungals or HIV	No antidote (Andexanet alfa in trials) ^e
Apixaban	Direct factor Xa inhibitor	Bleeding	Oral	No antidote (PER977 in trials) ⁵
Edoxaban	Direct factor Xa inhibitor	Bleeding	Oral	No antidote (PER977 in trials) ⁵

Original data from:

^aPatrono C, Collier B, FitzGerald GA, et al. Platelet-active drugs: the relationships among dose, effectiveness, and side effects: the Seventh ACCP Conference on Antithrombotic and Thrombolytic Therapy. *Chest*. 2004;126(3_suppl):234S–264S.

^bHirsh J, Raschke R. Heparin and low-molecular-weight heparin: the Seventh ACCP Conference on Antithrombotic and Thrombolytic Therapy. *Chest*. 2004;126(3_suppl):188S–203S.

^cWeitz JI, Hirsh J, Samama MM. New anticoagulant drugs: the Seventh ACCP Conference on Antithrombotic and Thrombolytic Therapy. *Chest*. 2004;126(3_suppl):265S–286S.

^dAgno W, Gallus AS, Wittkowsky A, et al. Oral anticoagulant therapy: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *CHEST*. 2012;141(2_suppl):e44S–e88S.

^eLeitch J, van Vlymen J. Managing the perioperative patient on direct oral anticoagulants. *Can J Anaesth*. 2017;64(6):656–672.

fracture and surgery can cause prolonged elevations with or without thromboembolic disease. A negative D-dimer excludes DVT or PE in patients with low probability of thromboembolic disease.³³ The first-line test to confirm a diagnosis of an acute PE is a multidetector spiral (helical) CT pulmonary angiography, where PE causes an intravascular filling defect in a pulmonary artery that occludes all or part of the vessel (Fig. 14.3A). The helical CT's excellent sensitivities and specificities of greater than 95% have raised concerns about overdiagnosing clinically unimportant PEs. The growing use of spiral CT has been linked to an increase in both the diagnosis of PEs and bleeding complications but has not been associated with a decrease in overall mortality rate from PE.^{34,35} Most recent guidelines recommend withholding anticoagulation for smaller subsegmental pulmonary embolisms (SSPE) with low risk for recurrent VTE.³⁶ Ventilation-perfusion scintigraphy scans (V/Q scans) lack the sensitivity and specificity of the spiral CT and are now a second line study. A negative, or low, probability excludes significant PE, but V/Q scans are best reserved for patients with contraindications to radiographic dye.³⁷

THROMBOEMBOLIC DISEASE TREATMENT

Prompt treatment prevents progression of thrombosis and minimizes mortality from embolic disease. Hematologists and internal medicine specialists are often the most expert in the management

of VTE; however, knowledge of current guidelines can be helpful to provide reassurance to patients and to expedite care when needed (Table 14.4). Patients who sustain their first DVT provoked by surgery are treated with 3 months of anticoagulation therapy with novel oral anticoagulants (NOACs; examples: factor Xa inhibitors or direct thrombin inhibitors), vitamin K antagonists (VKA = warfarin), or low-molecular-weight heparin.³⁸ Early mobilization after the diagnosis of DVT is encouraged without fear of dislodging the thrombus and creating an embolism. Studies using VQ scans have demonstrated no decrease in PE when bed rest was used after DVT diagnosis. Pain and swelling resolves more quickly when early ambulation is encouraged.^{39–41} Graded compression stockings (GCSs) may help with symptoms of leg swelling, but they do not relieve symptoms of pain better than placebo⁴² and do not prevent postthrombotic syndrome after a first DVT.⁴³ Early discharge and home management is possible and is a cost-effective approach for many DVTs and PEs.^{39,44} Intravenous vena cava filter use is generally discouraged but may play a role when there is a contraindication to or complication of anticoagulation therapy.^{38,39} Finally, after experiencing a second separate episode of VTE, a patient may benefit from receiving anticoagulation for an indefinite period.

Isolated venous thrombosis of the calf veins is not an uncommon issue in orthopedic patients. As ultrasonographers develop better tools and skills, this far distal pathology of the smallest

TABLE 14.3 Rates of Symptomatic Thromboembolism in Sports Medicine

RATES OF SYMPTOMATIC THROMBOEMBOLISM		
Procedure	sDVT (%)	PE (%)
All hospital admission ^a	.048–.07	.023–.03
Major ortho: THA, TKA, HFS		
In 2 weeks no prophylaxis ^b	1.8	1
In 35 days no prophylaxis ^b	2.8	1.5
In hospital with prophylaxis ^c	.26–.8	.14–.35
In 35 days with prophylaxis ^b	.45	.20
Knee arthroscopy ^d	.25–9.9	.028–.17
ACL reconstruction ^e	.3	.8
Hip arthroscopy ^f	0–3.7	0
Shoulder arthroscopy ^g	.01–.26	.01–.21
Shoulder fracture ^h	0	.2
Shoulder arthroplasty ^g	.19–.2	.1–.4
Elbow arthroplasty ^j		.25
Foot and ankle surgery ⁱ	0–.22	.02–.15
Ankle fracture ^k	.05–2.5	.17–.47
Ankle arthroscopy ^l	0	0

Original data from:

^aWhite RH. The epidemiology of venous thromboembolism. *Circulation*. 2003;107(90231):1-4-8.

^bFalck-Ytter Y, Francis CW, Johanson NA, et al. Prevention of VTE in orthopedic surgery patients: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2_suppl):e278S–e325S.

^cJanuel J, Chen G, Ruffieux C, et al. Symptomatic in-hospital deep vein thrombosis and pulmonary embolism following hip and knee arthroplasty among patients receiving recommended prophylaxis: a systematic review. *JAMA*. 2012;307(3):294–303.

^dHetsroni I, Lyman S, Do H, et al. Symptomatic pulmonary embolism after outpatient arthroscopic procedures of the knee: the incidence and risk factors in 418,323 arthroscopies. *J Bone Joint Surg Br*. 2011;93(1):47–51.

^eJameson SS, Dowen D, James P, et al. Complications following anterior cruciate ligament reconstruction in the English NHS. *Knee*. 2012;19(1):14–19.

^fBushnell BD, Anz AW, Bert JM. Venous thromboembolism in lower extremity arthroscopy. *Arthroscopy*. 2008;24(5):604–611.

^gOjike NI, Bhadra AK, Giannoudis PV, et al. Venous thromboembolism in shoulder surgery: a systematic review. *Acta Orthop Belg*. 2011;77(3):281–289.

^hJameson SS, James P, Howcroft DW, et al. Venous thromboembolic events are rare after shoulder surgery: analysis of a national database. *J Shoulder Elbow Surg*. 2011;20(5):764–770.

ⁱDuncan SF, Sperling JW, Morrey BF. Prevalence of pulmonary embolism after total elbow arthroplasty. *J Bone Joint Surg Am*. 2007;89(7):1452–1453.

^jJameson SS, Augustine A, James P, et al. Venous thromboembolic events following foot and ankle surgery in the English National Health Service. *J Bone Joint Surg Br*. 2011;93(4):490–497.

^kSooHoo NF, Eagan M, Krenek L, et al. Incidence and factors predicting pulmonary embolism and deep venous thrombosis following surgical treatment of ankle fractures. *Foot Ankle Surg*. 2011;17(4):259–262.

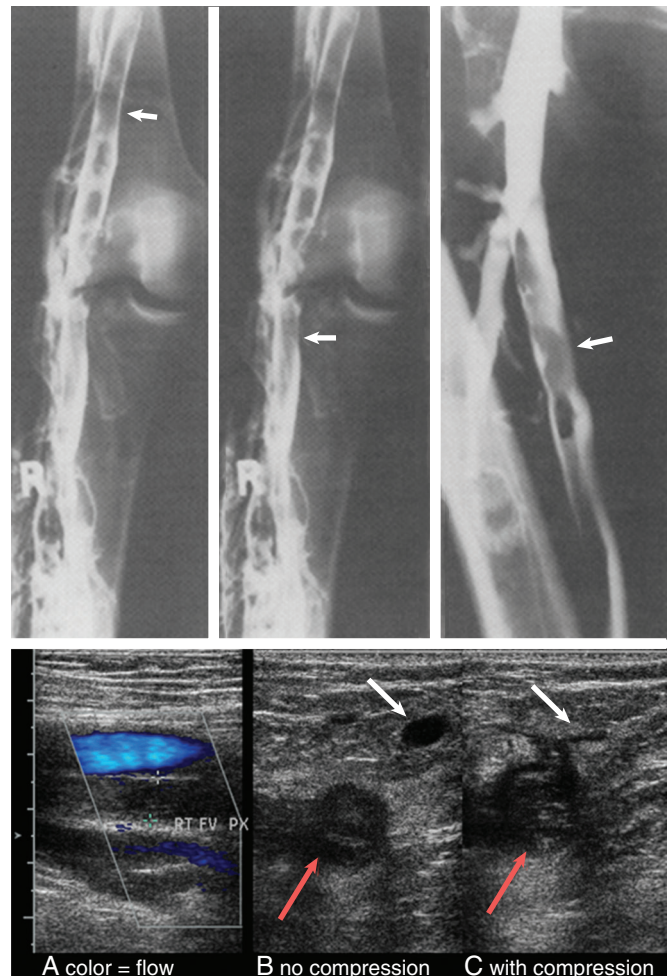


Fig. 14.2 Upper panel, The “gold standard” diagnosis for deep vein thrombosis: Intraluminal filling defects seen on two or more views of venogram. Left two images at the knee; right image at the hip. Lower panel, A color duplex ultrasound of femoral vessels with a venous clot in the deeper vessel. (B) demonstrates cross section ultrasound without compression and (C) demonstrates the same view with compression. Red arrows demonstrate deep vein with venous clot that fails to collapse with compression. White arrows demonstrate the normal more superficial vein collapses (no clot). (Upper panel, From Jackson JE, Hemingway AP. Principles, techniques and complications of angiography. In: Grainger RG, ed. *Grainger & Allison’s Diagnostic Radiology: A Textbook of Medical Imaging*. 4th ed. Philadelphia: Churchill Livingstone, 2011. Lower panel, Original images courtesy Austin Radiological Association and Seton Family of Hospitals.)

leg veins is more easily diagnosed. Experts suggest that this can be managed with surveillance with repeat duplex ultrasound at 2 weeks to look for progression of the clot. Treatment may be warranted if the patient has severe symptoms, if the clot is large (>5 cm, involves multiple veins, or is >7 mm in diameter), or if the patient is considered high risk (unprovoked, history of prior VTE or cancer).³⁸

SPORTS ISSUES, RETURN TO PLAY, AND TRAVEL

Spontaneous DVTs and PEs in athletes with and without thrombophilia can occur.^{45–50} Comparison studies show a risk reduction

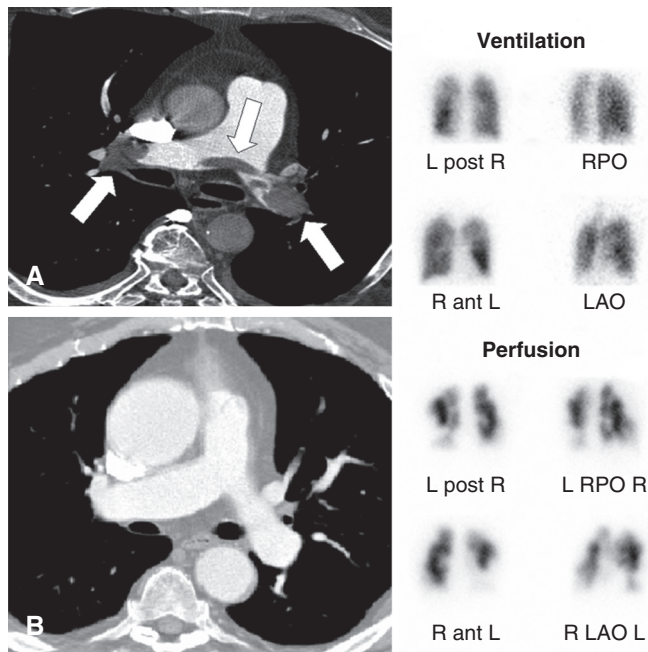


Fig. 14.3 Left image, Helical computed tomography pulmonary angiography. (A) Patient with symptoms of syncope and hypoxia. Large clots (arrows) are present at the bifurcation of the main pulmonary artery and extend into the left and right pulmonary arteries. (B) Normal scan in a different patient at same level. Right image, High-probability pulmonary V/Q scan: Ventilation scan above demonstrates full lung fields. Perfusion scan below shows multiple areas lacking tracer. (Original image courtesy Austin Radiological Association and Seton Family of Hospitals.)

in thromboembolic disease with activity⁵¹ and sports participation.⁵² Although DVTs usually affect athletes in the lower extremity, those who present with upper extremity pain, swelling, feeling of heaviness or dilated upper arm veins, the diagnosis of axillary-subclavian vein thrombosis or “effort thrombosis” (Paget-Schroetter syndrome) should be entertained.⁵³ Sports that involve shoulder abduction and extension (wrestling, gymnastics, weightlifting, throwing, and swimming) may have increased incidence in patients with anatomic abnormality of their thoracic outlet (cervical rib, congenital bands, hypertrophy of scalenus tendons, and/or abnormal insertion of the costoclavicular ligament). Upper extremity duplex ultrasound may help diagnosis, and thoracic outlet decompression may optimize their care.^{54,55}

Athletes with DVT should be encouraged to start a gradual return to their usual daily activities when they begin anticoagulation therapy. A formal return-to-training program with progressively increasing intensity can be undertaken shortly thereafter, if the athlete is closely monitored for recurrence of VTE. Traditionally, athletes have been prohibited from contact or collision sports until anticoagulation therapy has been completed; however, the NOACs may allow protocols that allow earlier participation.^{56,57}

Athletes involved in international competition may be at increased risk for “traveler’s thrombosis” (also known as “economy-class syndrome” or “rail coach syndrome”⁵⁸) when facing travel durations longer than 4 hours.⁵⁹ Young elite athletes who are at their peak physical condition are unlikely to experience significant health compromise; however, subclinical

thromboembolic disease could potentially jeopardize performance.^{1,60} Multiple studies have made an association between taller individuals and increased risk of a first or recurrent venous thromboembolism.^{61–64} Injuries during competition and their treatment may put athletes at increased risk on their return journey. Prolonged air travel before surgery may also increase the risk of perioperative VTE.^{65–67} The risk of VTE is related to a person’s risk factors and the duration of the flight. The cause of the increased risk is likely related to a decrease in vascular flow of immobility; however, dehydration, decreased cabin pressure, and relative hypoxia may also play a role. The absolute risk of fatal PE is very low (2.57 per 1 million flights that last longer than 8 hours),⁶⁸ but air travel lasting longer than 8 hours carries eight times the risk of fatal PE in nontravelers.^{69,70} Most patients who experience travel-related thrombosis have one or more other risk factors. Potential ways to help reduce the risk of VTE are by avoiding constrictive clothing, staying adequately hydrated, performing calf-stretching exercises, and taking frequent walks in the cabin.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

van Adrichem RA, Nemeth B, Algra A, et al. Thromboprophylaxis after knee arthroscopy and lower-leg casting. *N Engl J Med*. 2017;376(6):515–525.

Level of Evidence:

II

Summary:

A randomized controlled trial of patients important to sports medicine doctors looking at clinically relevant thromboembolic disease in both knee arthroscopy and cast immobilized patients. After studying more than 3000 patients, the conclusion is that chemical prophylaxis was not effective in preventing symptomatic VTE.

Citation:

Kahn SR, et al. Compression stockings to prevent post-thrombotic syndrome: a randomised placebo-controlled trial. *Lancet*. 2014;383(9920):880–888.

Level of Evidence:

II

Summary:.

A multicenter randomized placebo-controlled trial of active versus placebo elastic compression stockings used for 2 years to prevent postthrombotic syndrome (PTS) after a first proximal DVT. Elastic compression hose did not prevent PTS after DVT, and their routine use was not supported.

Citation:

Falck-Ytter Y, et al. Prevention of VTE in orthopedic surgery patients: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(suppl 2):e278S–e325S.

Level of Evidence:

II

TABLE 14.4 Treatment of Acute Venous Thromboembolism Disease

Phases of anticoagulation				
Initial (0 to ~7 days) Long-term (~7 days to ~3 months) Extended (~3 months to indefinite)				
Parenteral Xa inhibitors, Dabagitan or Vit K				
Drug		Dose	Duration	Monitoring
Parenteral	IV heparin+	80 U/kg IV bolus 18 U/kg/h IV	≥5 days	aPTT: 1.8–2.5 × control or Anti-Xa activity: 0.3–0.7 U/mL
	SC heparin+	333 U/kg SC then 250 U/kg SC BID	≥5 days	aPTT: 1.5–2.5 × control QAM 6 h after am dose
	LMWH–	1 mg/kg BID SC 2 mg/kg QD SC	≥5 days	None—unless RI or pregnancy Anti-Xa activity: 0.6–1.0 I U/mL
	Fondaparinux–	7.5 mg SC QD 5 mg if <50 kg 10 mg if >100 kg	3 mos+	None
Oral	Rivaroxaban–#	10 mg PO QD		
	Apixaban	5 mg PO BID	3 mos+	None
	Edoxaban	60 mg PO QD		
	Dabigatran	150 mg PO QD	3 mos+	None (aPTT/thrombin clotting time)
	VKA	5–10 mg PO QD Adjust dose	3 mos+	INR: 2–3 QAM

+ preferred for patients with severe renal impairment
 – avoid in patients with marked renal impairment (<30 mL/min)
 # avoid in patients on azole (antifungals) and HIV protease inhibitors

Special Notes:
Bed rest: Not recommended, early mobilization for both DVT patients
Distal DVT: Repeat US in 2 weeks, only Rx if severe symptoms or ↑ risk: ≥5 cm long, ≥7 mm diameter, cancer, Hx of VTE, inpatient
Home DVT Rx: If home circumstances adequate (+phone, family or friends, near hospital)
Early PE D/C: If home circumstances adequate (+phone, family or friends, near hospital)
Subsegmental PE: If proximal LE veins duplex negative, clinical surveillance ok
GCS: Graded compression stockings *NOT* recommended for preventing VTE or postthrombotic syndrome
IVC filters: Only if contraindication to anticoagulation (May resume anticoagulation if bleeding risks resolve)
Superficial VT: If severe symptoms, then 45 days of prophylactic anticoagulation, if not serial duplex looking for extension
Upper extremity: Axillary or more proximal veins get anticoagulation (not GCS)
Controversial: In the postoperative arthroplasty situation, consider thrombectomy and embolectomy before thrombolytics

Duration of treatment: 3 mos use same drug.

DVT, Deep vein thrombosis; GCS, Graded compression stocking; IVC, intravena caval; VKA, vitamin K antagonists; VTE, venous thromboembolism disease.

Graphic modified from and original data from Kearon C, Akl EA, Comerota AJ, et al. Antithrombotic therapy for VTE disease: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2_suppl):e419S–e494S; Holbrook A, Schulman S, Witt DM, et al. Evidence-based management of anticoagulant therapy: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2_suppl):e152S–e184S; and Garcia DA, Baglin TP, Weitz JI, et al. Parenteral anticoagulants: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2_suppl):e24S–e43S.

Summary:

Chest physician's guidelines summarize a multispecialty consensus of the literature that is updated. This comprehensive guideline covers many topics rarely addressed in orthopedic literature. Optimal strategies for thromboprophylaxis after major orthopedic surgery include pharmacologic and mechanical approaches.

Citation:

Jacobs JJ, Mont MA, Bozic KJ, et al. Prevent venous thromboembolic disease in patients undergoing elective hip and knee arthroplasty: evidence-based guideline and evidence report; September 23, 2011. *AAOS Evidence-Based Clinical Practice Guidelines*. 1-861. http://www.aaos.org/Research/guidelines/VTE/VTE_full_guideline.pdf.

Level of Evidence:

III

Summary:

Generated by an AAOS physician Work Group and experts in systemic reviews, this AAOS clinical practice guideline is based on an assessment of the highest-level research available and makes recommendations summarized in the chapter. The full guideline available online shares the research details in depth.

Citation:

Grabowski G, Whiteside WK, Kanwisher M. Venous thrombosis in athletes. *J Am Acad Orthop Surg*. 2013;21(2):108–117.

Level of Evidence:

V

Summary:

This excellent review article in the JAAOS specifically summarizes the current literature on venous thrombosis in athletes and walks the orthopedic surgeon from the basic science level of DVT and completes the story by reviewing the prevention and management of DVTs.

Citation:

Tornetta P, Bogdan Y. Pulmonary embolism in orthopaedic patients: diagnosis and management. *J Am Acad Orthop Surg*. 2012;20(9):586–595.

Level of Evidence:

V

Summary:

Orthopedic surgeons will have a patient present with signs and symptoms concerning for a pulmonary embolism at some point during their careers. This review article does a great job scanning and summarizing the current scientific data and practices of diagnosing and managing those patients with a pulmonary embolism.

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